CNNs for Image Classification: A Comparative Study on CIFAR-10 Using ResNet-18, AlexNet, and SimpleCNN

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Abstract

In this study, I have explored the performance difference between Convolutional Neural Network (CNN) architectures for image classification using the CIFAR-10 dataset, a well-known benchmark in computer vision consisting of 60,000 images across 10 classes. I have implemented and rigorously compared three architectures: ResNet-18, AlexNet, and a custom-designed SimpleCNN. Each model has been trained from scratch, and I have applied a refined hyperparameter tuning process to ResNet-18. This involves optimizing learning rates, batch sizes, and selecting between the SGD and Adam optimizers. I have used techniques such as early stopping and learning rate scheduling to enhance model performance and prevent overfitting. My results show that ResNet-18 outperforms the other models, achieving a validation accuracy of 78.9% in basemodel of ResNet-18 and a final test accuracy of 80.5% after hyperparameter tuning, demonstrating its robustness and generalization capabilities. AlexNet and SimpleCNN, while simpler, provide insights into the trade-offs between model complexity and performance, achieving test accuracies of 71.6% and 72.5%, respectively. I have presented a detailed analysis of performance metrics, including accuracy and loss curves. These findings underscore the importance of hyperparameter selection and architectural choices in deep learning models.

1. Introduction

Image classification is a core task in computer vision, involving the categorization of images into predefined classes using computer algorithms. It has widespread applications, from facial recognition to medical image analysis. In this study, I have employed deep learning techniques, specifically Convolutional Neural Networks (CNNs), to address the image classification problem on the CIFAR-10 dataset.

CNNs have transformed image classification due to their ability to automatically learn hierarchical features from raw data. Inspired by biological neural networks, CNNs use interconnected layers to extract complex patterns. The foundational work by LeCun on LeNet [6] laid the groundwork for CNN development, with Krizhevsky's AlexNet [5] achieving groundbreaking results on ImageNet, inspiring further advancements in CNN architectures.

My primary goal in this research is to implement and compare three CNN architectures: ResNet-18, AlexNet, and a custom SimpleCNN. ResNet-18's residual learning framework effectively trains deep networks by addressing the vanishing gradient issue. AlexNet, though simpler, highlights key CNN design principles, while SimpleCNN serves as a baseline to explore the trade-offs between complexity and performance.

I have trained and evaluated each model rigorously, applying techniques like early stopping and learning rate scheduling to enhance performance and reduce overfitting. Additionally, I have performed hyperparameter tuning to investigate the effects of different learning rates, batch sizes, and optimizers. This comprehensive analysis aims to identify the best-performing architecture and provide insights into CNN effectiveness on the CIFAR-10 dataset.

2. Background

Convolutional Neural Networks (CNNs) have become fundamental in image classification, thanks to their ability to automatically learn hierarchical features from raw image data. I have explored several significant advancements in CNN research to guide this study.

Kaiming He [2] introduced Residual Networks (ResNets), which address the vanishing gradient problem through shortcut connections, enabling the effective training of deep networks. I have selected ResNet-18 for its balance between depth and computational efficiency, making it suitable for the CIFAR-10 dataset.

Howard [3] presented MobileNets, which reduce computational cost using depthwise separable convolutions. Al-

though my focus is not on MobileNets, this work highlights the importance of efficient architecture design, especially for resource-constrained tasks.

Smith [10] emphasized the significance of optimal learning rates and batch sizes, introducing cyclical learning rates to accelerate convergence and improve model performance. Inspired by this, I have systematically tuned hyperparameters to enhance model accuracy and generalization.

Perez and Wang [8] demonstrated that data augmentation, including transformations like rotations and flips, effectively mitigates overfitting. I have applied similar techniques to improve the robustness of my models.

Ioffe and Szegedy [4] proposed Batch Normalization, a method that stabilizes and accelerates training by normalizing layer inputs, allowing higher learning rates and better performance. I have incorporated this approach to enhance the training stability of my CNN models.

These studies collectively inform my implementation and comparison of ResNet-18, AlexNet, and SimpleCNN, emphasizing how architectural design and optimization techniques impact performance on image classification tasks.

3. Methodology

3.1. Dataset

I have used the CIFAR-10 dataset, a widely recognized benchmark for image classification tasks. The dataset consists of 60,000 color images, each of size 32x32 pixels, evenly distributed across 10 classes, including airplanes, automobiles, birds, cats, and more. The data is divided into a training set of 50,000 images and a test set of 10,000 images, ensuring a balanced representation across classes. Due to the small size of the images and the complex nature of the classification task, it is crucial to employ effective preprocessing techniques to maximize model performance.

3.2. Preprocessing

To prepare the data for training, I applied several essential preprocessing steps. Firstly, I normalized the pixel values of the images to the range [0, 1] by dividing each pixel value by 255. Normalization is a standard practice in deep learning, as it helps to accelerate convergence and ensure consistent input distributions for the models.

I then split the original test set into two equal parts, resulting in a validation set and a test set, each containing 5,000 images. This split allows me to monitor model performance during training and adjust hyperparameters accordingly. To ensure reproducibility, I fixed the random seed when performing this split.

I decided not to implement data augmentation for two main reasons. Firstly, the CIFAR-10 dataset is already sufficiently large and diverse for meaningful model comparison without the need for artificially increasing the dataset size. This dataset size allowed me to effectively evaluate and compare different CNN architectures. Secondly, given the computational and time constraints of the project, I prioritized focusing on model architecture and hyperparameter tuning. By doing so, I was able to allocate resources efficiently and still gain valuable insights into the performance of various CNN models on the CIFAR-10 dataset.

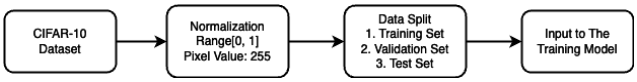


Figure 1. Data Preprocessing Flow

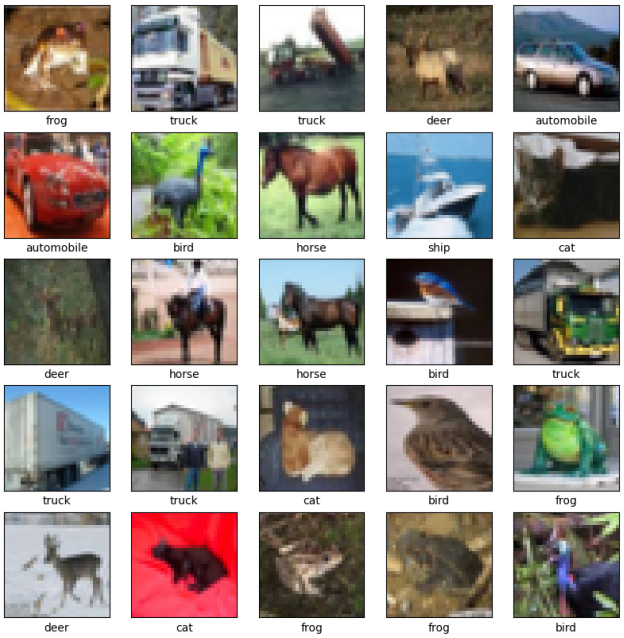


Figure 2. Sample Image of CIFAR-10 Dataset

3.3. Exploratory Data Analysis (EDA)

To better understand the CIFAR-10 dataset, I performed Exploratory Data Analysis (EDA). Figure 2 shows a 5x5 grid of random sample images from the training set, highlighting the diversity across the 10 classes, such as airplanes, automobiles, and animals. This visualization provides a clear overview of the image content and variation.

I also analyzed the class distribution, shown in Figure 3. Each class contains an equal number of 6,000 samples, ensuring a balanced dataset. This uniform distribution prevents class imbalance issues, allowing for unbiased model training and evaluation. The EDA confirms that the dataset is well-suited for deep learning tasks without requiring additional balancing.

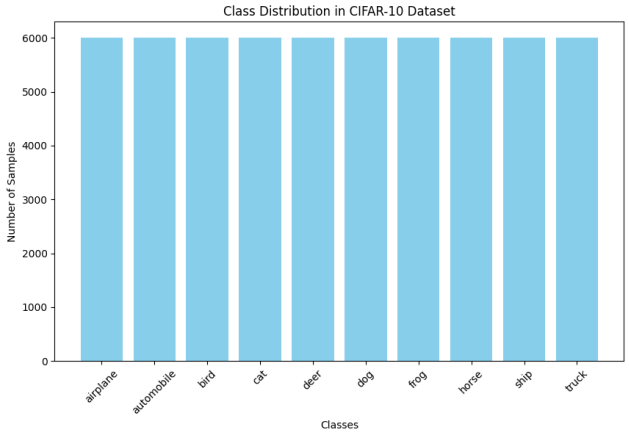


Figure 3. Class Distribution of CIFAR-10 Dataset

3.4. Deep Learning Models

3.4.1 ResNet-18

I have constructed the ResNet-18 model, which is part of the ResNet family known for addressing the vanishing gradient problem in deep networks. The core concept is the use of *skip connections* or *identity mappings*, allowing the model to learn residual functions, $\mathcal{F}(x) + x$, instead of traditional unreferenced mappings $\mathcal{F}(x)$.

The residual block, depicted in Figure 4, forms the building block of the ResNet-18 architecture. The block includes two weight layers, with a shortcut connection that bypasses these layers. Mathematically, the output of a residual block can be expressed as:

$$y = \sigma(\mathcal{F}(x) + x) \tag{1}$$

where:

- x is the input to the residual block.
- $\mathcal{F}(x)$ is the learned residual function consisting of two convolutional layers followed by batch normalization and ReLU activation.
- σ denotes the ReLU non-linearity applied to the output.

The ResNet-18 model is composed of:

1. An initial convolutional layer with batch normalization and ReLU activation.
2. Four stages of residual blocks, with each stage increasing the number of filters while down-sampling the spatial dimensions (where necessary) using strided convolutions.
3. A global average pooling layer followed by a dense layer with a softmax activation for classification.

The model is defined mathematically as follows:

$$\text{Conv1}(x) = \sigma(\text{BN}(\text{Conv}(x))) \tag{2}$$

$$\text{Output} = \text{GlobalAveragePooling} \rightarrow \text{Dense(Softmax)} \tag{3}$$

These equations outline how the architecture processes input images, with each block refining the features learned by the model.

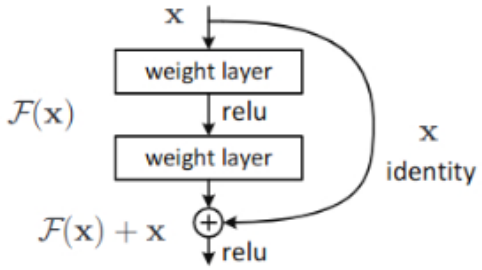


Figure 4. Illustration of a residual block with skip connections in ResNet-18. The input x is added to the output of two weight layers, enabling the learning of residual mappings [9]

3.4.2 AlexNet

I implemented the AlexNet architecture from scratch to evaluate its performance on the CIFAR-10 dataset. AlexNet features five convolutional layers followed by three fully connected layers. Figure 5 shows the model's structure, utilizing max-pooling layers to reduce spatial dimensions and ReLU activations to introduce non-linearity.

The convolutional operation is defined as:

$$\text{Conv}(X, W, b) = (X * W) + b, \tag{4}$$

where X is the input, W the filter, $*$ the convolution operation, and b the bias. ReLU activation is expressed as:

$$\text{ReLU}(x) = \max(0, x), \tag{5}$$

introducing non-linearity to learn complex patterns. Max-pooling reduces feature map dimensions:

$$\text{MaxPool}(X) = \max_{i,j \in \text{window}} X_{i,j}. \tag{6}$$

Dropout regularization is used in fully connected layers to mitigate overfitting. The output layer applies softmax activation:

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}, \tag{7}$$

normalizing class scores into probabilities.

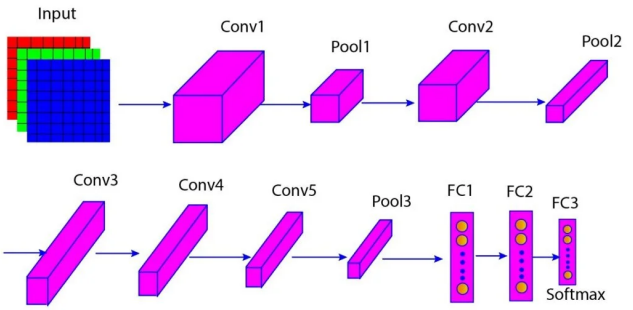


Figure 5. AlexNet architecture: five convolutional layers with max-pooling and three fully connected layers [7]

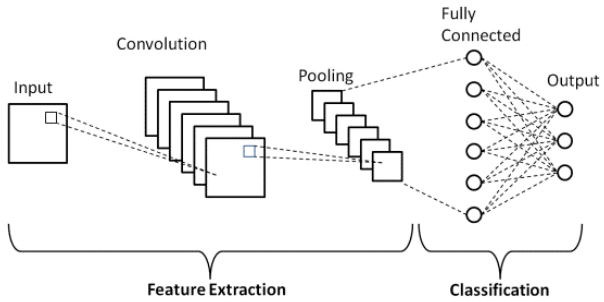


Figure 6. SimpleCNN architecture illustrating convolutional layers for feature extraction, followed by fully connected layers for classification [1]

I optimized the model with Adam (learning rate 0.001) and used sparse categorical cross-entropy loss. Callbacks like ReduceLROnPlateau and EarlyStopping were employed to enhance training stability. The model trained for 50 epochs with a batch size of 64, with performance tracked on a validation set.

3.4.3 SimpleCNN

I developed the SimpleCNN model to serve as a lightweight, baseline architecture for evaluating image classification performance on the CIFAR-10 dataset. The model, shown in Figure 6, consists of three convolutional layers, each followed by a max-pooling layer, and concludes with a dense network for classification.

The first convolutional layer applies 32 filters with a 3x3 kernel size, followed by a ReLU activation and a 2x2 max-pooling operation to reduce spatial dimensions. The next two convolutional layers progressively increase the number of filters to 64 and 128, using similar activation and pooling mechanisms. Finally, the feature maps are flattened and passed through a fully connected layer with 128 neurons and a ReLU activation, followed by dropout regularization to prevent overfitting. The final layer uses a softmax activation for classifying the input into one of the 10 categories.

To train the model, I employed the Adam optimizer with an initial learning rate of 0.001 and used the sparse categorical cross-entropy loss function. The training process was monitored using validation data, and callbacks such as ReduceLROnPlateau and EarlyStopping were implemented to optimize learning rates and halt training when the model performance plateaued. The model was trained for 50 epochs with a batch size of 64, ensuring efficient convergence and avoiding overfitting.

3.5. Hyperparameter Optimization

Hyperparameter tuning plays a critical role in optimizing the performance of deep learning models. I employed a grid search strategy to explore various hyperparameter configurations for the ResNet-18 model on the CIFAR-10 dataset. The goal was to identify the best combination of learning rate, batch size, and optimizer for maximum validation accuracy and minimal loss.

The hyperparameter grid was defined as follows:

- **Learning Rates**: $\{0.001, 0.0005\}$
- **Batch Sizes**: $\{64, 128\}$
- **Optimizers**: $\{\text{SGD, Adam}\}$

The model was trained using each combination of hyperparameters. I initialized variables to store the best validation accuracy, lowest validation loss, and the corresponding hyperparameters. For each configuration, I instantiated the ResNet-18 model and compiled it with the chosen optimizer and a learning rate. The loss function used was sparse categorical cross-entropy, and the accuracy metric was monitored throughout training.

To further stabilize and optimize the training process, I implemented two callbacks:

1. **EarlyStopping**: Monitored the validation loss and stopped training if there was no improvement for five consecutive epochs, restoring the best weights.

2. **ReduceLROnPlateau**: Reduced the learning rate by a factor of 0.5 if the validation loss did not improve for three epochs, with a minimum learning rate threshold of 1×10^{-6}.

Each model was trained for up to 15 epochs, and the training progress, including the current epoch number, was displayed for monitoring. I iterated through all completed epochs to evaluate the validation accuracy and loss. If a configuration achieved a higher validation accuracy or had a lower validation loss when accuracies were equal, I updated the best hyperparameters accordingly.

The best hyperparameters and their corresponding validation metrics were acquired at the end of the search. This systematic approach allowed me to fine-tune the model efficiently and enhance its performance on the validation set.

4. Experiments and Results

4.1. Evaluation Metrics

I evaluated ResNet-18, AlexNet, and SimpleCNN using several metrics on the validation set:

Model Accuracy	Training	Validation
ResNet-18	95.6%	78.9%
AlexNet	80.6%	71.6%
SimpleCNN	78.2%	72.5%

Table 1. Training and validation accuracies (in percentages) for ResNet-18, AlexNet, and SimpleCNN.

Model	Precision	Recall	F1-Score
ResNet-18	0.81	0.81	0.80
AlexNet	0.73	0.73	0.73
SimpleCNN	0.72	0.72	0.72

Table 2. Key performance metrics including precision, recall, and F1-score for ResNet-18, AlexNet, and SimpleCNN on the validation set.

- **Accuracy:** The ratio of correct predictions to the total number of samples, indicating the model's overall effectiveness.
- **Loss:** Sparse categorical cross-entropy loss, measuring the difference between true labels and predicted probabilities. Lower values indicate better performance.
- **Training and Validation Curves:** Plots of accuracy and loss over epochs to detect overfitting or underfitting.
- **Classification Report:** Precision, recall, and F1-score for each class, providing detailed performance metrics across CIFAR-10 categories.
- **Confusion Matrix:** Visual representation comparing actual and predicted class labels, helping identify misclassified instances.

These metrics offer a comprehensive framework for comparing model performance.

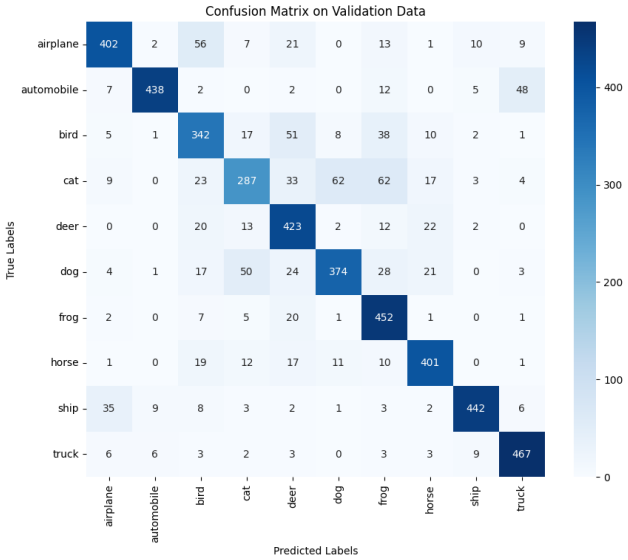


Figure 7. ResNet-18 Confusion Matrix

4.2. Model Performance

4.2.1 ResNet-18 Model Performance

Key insights from the ResNet-18 model evaluation are summarized as follows:

- **Training and Validation Accuracy:**
  - Training accuracy: 95.6%
  - Validation accuracy: 78.9%
- **Performance Observations:**
  - Significant gap between training and validation accuracy, indicating overfitting.
  - Strong performance on classes like "automobile" and "ship."
  - Challenges observed in accurately predicting classes like "cat" and "bird."
- **Confusion Matrix (Figure 7):**
  - Highlights areas of confusion between similar classes, such as "dog" and "cat."
- **Classification Report (Table 2):**
  - Overall weighted F1-score: 0.80
  - Precision and recall variations across classes, reflecting prediction imbalances.

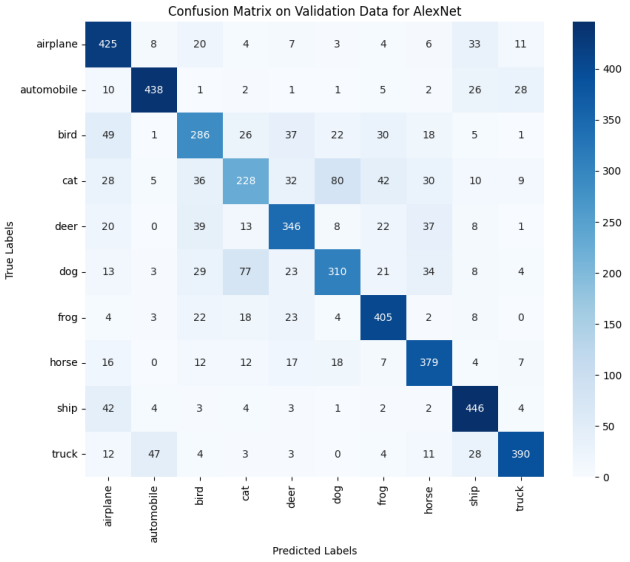


Figure 8. AlexNet Confusion Matrix

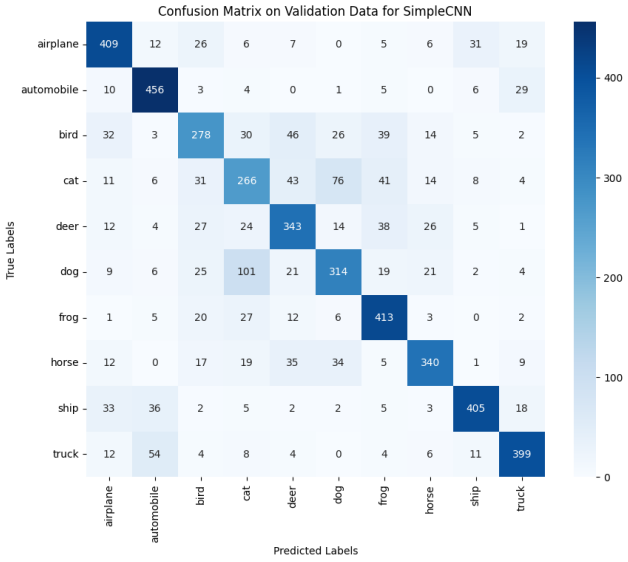


Figure 9. SimpleCNN Confusion Matrix

4.2.2 AlexNet Model Performance

Key insights from the AlexNet model evaluation are summarized as follows:

- **Training and Validation Accuracy:**
  - Training accuracy: 80.6%
  - Validation accuracy: 71.6%
- **Performance Observations:**
  - Moderate gap between training and validation accuracy, suggesting potential overfitting.
  - Strong performance on classes like "automobile" and "ship."
  - Weaker performance on classes such as "cat" and "bird."
- **Confusion Matrix (Figure 8):**
  - Visual representation of class-wise predictions, highlighting areas of misclassification.
- **Classification Report (Table 2):**
  - Overall weighted F1-score: 0.73
  - Detailed precision, recall, and F1-scores showcasing imbalances in class prediction quality.

4.2.3 SimpleCNN Model Performance

- **Training and Validation Accuracy:**
  - Training accuracy: 78.2%
  - Validation accuracy: 72.5%

Performance Analysis:

- **Accuracy and Loss Trends:**
  - The training accuracy steadily increased, reaching 78.2%, while the validation accuracy stabilized around 72.5%.
  - The training and validation loss curves indicate that the model converged but exhibited slight overfitting as the gap between the two losses increased towards the end of training.
- **Confusion Matrix (Figure 9):**
  - The model had strong performance in classes like "automobile" and "ship" but showed notable misclassifications in "cat" and "bird" categories.
- **Classification Report (Table 2):**
  - SimpleCNN achieved a weighted average precision, recall, and F1-score of 72%, suggesting a balanced performance across most classes, albeit with room for improvement in underperforming categories.

4.3. Comparison Analysis and Best Model Selection

Among the three models—ResNet-18, AlexNet, and SimpleCNN—ResNet-18 clearly outperformed the others with a training accuracy of 95.6% and a validation accuracy of 78.9%. The use of residual connections in ResNet-18

effectively addressed the vanishing gradient problem, enabling better feature extraction and generalization. Its loss curves indicated stable learning with minimal overfitting, making it the most robust architecture for this task.

AlexNet achieved a training accuracy of 80.6% and a validation accuracy of 71.6%, but it exhibited signs of overfitting, reflected in a wider gap between training and validation performance. Its simpler architecture struggled to capture complex features, limiting its effectiveness. SimpleCNN, with accuracies of 78.2% (training) and 72.5% (validation), served as a decent baseline but lacked the depth to handle more intricate patterns, resulting in higher validation loss.

In conclusion, ResNet-18 stood out as the best model, demonstrating superior accuracy, efficient training, and strong generalization, making it the optimal choice for CIFAR-10 classification.

4.4. Hyperparameter Tuning Outcomes

To optimize the performance of ResNet-18, I conducted a hyperparameter tuning process. The best combination of hyperparameters, as shown in Table 3, resulted in a learning rate of 0.0005, a batch size of 64, and the Adam optimizer. This configuration achieved a validation accuracy of 85.4% and a lowest validation loss of 0.7046.

Using these optimal hyperparameters, I trained the model for additional epochs to enhance its performance further. Figure 10 and Figure 11 illustrates the accuracy and loss curves for this training phase, showing improved model generalization. The training process incorporated an early stopping mechanism to prevent overfitting, with the model ultimately achieving a validation accuracy of 80.4%.

Hyperparameter	Value
Learning Rate	0.0005
Batch Size	64
Optimizer	Adam

Table 3. Best hyperparameters obtained through the tuning process for ResNet-18.

The code snippet used for the final training phase is shown below. This configuration leverages the identified optimal settings to achieve robust model performance.

4.5. Test Set Evaluation

To assess the generalization performance of the best-tuned ResNet-18 model, I evaluated it on the test set finally. The model achieved a final test loss of 0.622 and an accuracy of 80.5%, reflecting a strong performance in classifying the CIFAR-10 dataset. However, error analysis through the confusion matrix reveals areas for potential improvement. For instance, the model struggles to differentiate between certain classes, such as cats and dogs, or ships and trucks, which may be due to similarities in their features or background contexts.



Figure 10. Training and validation loss curves for ResNet-18 with the best hyperparameters

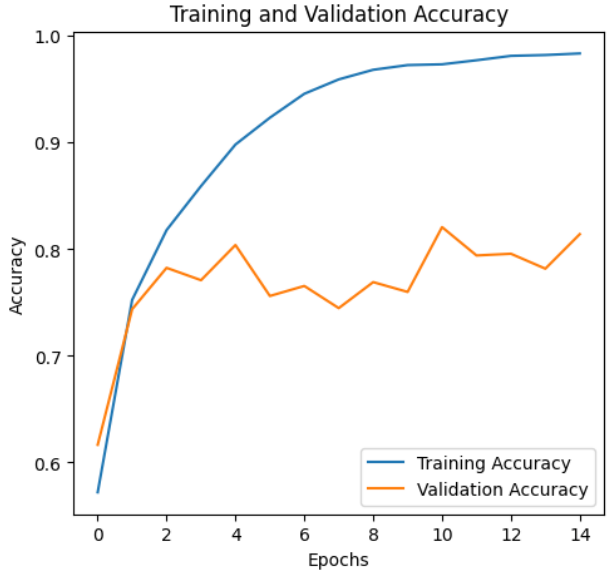


Figure 11. Training and validation accuracy curves for ResNet-18 with the best hyperparameters

The classification report provides additional insights, highlighting high precision and recall scores for classes like *automobile* and *truck*, while more challenging categories, such as *cat* and *bird*, have lower scores. The macro average and weighted average metrics both confirm consistent performance across most classes. These findings suggest that

although ResNet-18 performs well overall, optimizing feature learning for visually similar classes remains a key area for further improvement.

5. Discussion

The comparative analysis of ResNet-18, AlexNet, and SimpleCNN on the CIFAR-10 dataset provides several key insights into the strengths and limitations of these architectures. ResNet-18 demonstrated superior performance with a final test accuracy of 80.5%, attributed to its use of residual connections that effectively address the vanishing gradient problem. This architecture showed strong generalization capabilities and a well-balanced precision and recall across most classes, making it suitable for real-world applications where robustness is critical. However, there were still challenges in distinguishing visually similar classes, indicating room for further improvement in feature extraction techniques. The major design choice of using ResNet-18 highlights the importance of depth and skip connections for efficient learning, supported by the experimental evidence of improved accuracy.

In contrast, AlexNet and SimpleCNN achieved lower accuracies of 71.6% and 72.5%, respectively. AlexNet's simpler structure limited its ability to capture complex features, while SimpleCNN served as a baseline model, highlighting the trade-offs between simplicity and performance. These models provide a foundation for understanding the impact of architectural depth and complexity on classification tasks. The findings emphasize the importance of choosing the right architecture and optimizing hyperparameters to achieve better generalization. Future work could explore data augmentation, transfer learning, or experimenting with more sophisticated models to address the identified limitations and further enhance classification accuracy, guided by our experimental insights.

6. Conclusion and Future Work

This study presented a comparative evaluation of three convolutional neural network architectures—ResNet-18, AlexNet, and SimpleCNN—on the CIFAR-10 image classification task. ResNet-18 emerged as the most effective model, achieving a test accuracy of 80.5%, which underscores the efficacy of residual connections in deep networks. AlexNet and SimpleCNN, while offering simpler implementations, demonstrated lower accuracies, highlighting the impact of architectural complexity on performance. These findings contribute to understanding the trade-offs between model depth, feature learning capacity, and generalization.

Future research could focus on enhancing model performance through data augmentation techniques, exploring transfer learning with pre-trained models, and implementing more sophisticated architectures such as DenseNet or EfficientNet. Additionally, employing advanced regularization methods and experimenting with different learning rate schedules may yield further improvements. Extending this work to larger and more diverse datasets can also provide insights into the models' robustness and scalability.

Code

The codebase and data are made available on GitHub to facilitate reproducibility and further exploration. https://github.com/ruhinb/CNNs-for-Image-Classification-A-Comparative-Study-on-CIFAR-10-Using-ResNet-18-AlexNet-and-SimpleCNN/tree/main

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